Tanaka Certificates

July 1, 2026

Abstract

In these notes, I am building up the intuition and theory behind supermartingale certificates, with relaxed assumptions on the certificate function V. In particular, it is explored what happens when V is not twice continously-differentiable, but instead piecewise twice continuously-differentiable.

1 Piecewise linear certificates in dimension one

Consider the following setup:

Setup

Let:

1. $dX_t = f(X_t)dt + g(X_t)dW_t$ be a one-dimensional SDE
2. $V(x)$ be a piecewise linear function such that $V(x) = a_i x + b_i$, where $x \in (x_i, x_{i+1})$.
3. $V(x)$ is continuous at breakpoints: $V(x_i^+) = V(x_i^-) = V(x_i)$
4. as a consequence, $V'_-(x_i) = a_{i-1}$, $V'_+(x_i) = a_i$

Now we will consider what conditions on V are sufficient to guarantee that $V(X_t)$ is a supermartingale.

Hypothesis

Does concavity of $V(x)$ (along with an interior / drift term condition) guarantee that $V(X_t)$ is a supermartingale?

To answer this, let's apply the Itô-Tanaka formula to $V(X_t)$:

$$
\begin{aligned}
V(X_t) = V(X_0) + & \int_0^t V'(X_s)f(X_s)ds && (1)\\
+ & \int_0^t V'(X_s)g(X_s)dW_s && (2)\\
+ & \frac{1}{2}\sum_{i=1}^{n-1}\left(V'_+(x_i) - V'_-(x_i)\right)L_t^{x_i}(X) && (3)\\
& && (4)
\end{aligned}
$$

We will focus on these terms:

1. $D_t = \int_0^t V'(X_s)f(X_s)ds$, the drift term
2. $M_t = \int_0^t V'(X_s)g(X_s)dW_s$, the diffusion term (stochastic integral)
3. $K_t = \frac{1}{2}\sum_{i=1}^{n-1}\left(V'_+(x_i) - V'_-(x_i)\right)L_t^{x_i}(X)$, the kink term

we need to prove that $D_t + M_t + K_t$ is a supermartingale.

Some natural assumptions, which are sufficient to guarantee that $V(X_t)$ is a supermartingale, are the following:

1. $f(x)V'(x) \leq 0$ for all $x \in (x_i, x_{i+1})$. This is because of the following:

Fact

If $f(X)V'(X) \leq 0$, then the drift term D_t is a supermartingale.

Proof: We want to prove $\mathbb{E}[D_t|\mathcal{F}_s] \leq D_s$ for $s \leq t$.

$$\mathbb{E}[D_t|\mathcal{F}_s] = \mathbb{E}\left[\int_0^t V'(X_u)f(X_u)du \middle| \mathcal{F}_s\right] \tag{5}$$
$$= \int_0^s V'(X_u)f(X_u)du + \mathbb{E}\left[\int_s^t V'(X_u)f(X_u)du \middle| \mathcal{F}_s\right] \tag{6}$$
$$\leq D_s + 0 = D_s \tag{7}$$

2. M_t is a martingale. Assuming $V(x)$ has finitely many pieces (as is the case for a neural network with ReLU activation functions), it is enough to impose a square-integrability condition on $g(x)$[1]
3. $V'_+(x_i) - V'_-(x_i) \leq 0 \iff a_{i-1} \geq a_i$, i.e. the **concavity condition**. This is because of the following:

Fact

If $V'_+(x_i) - V'_-(x_i) \leq 0$ and $L_t^{x_i}(X) \geq 0$, then the kink term K_t is a supermartingale.

Proof: We want to prove $\mathbb{E}[K_t|\mathcal{F}_s] \leq K_s$ for $s \leq t$.

$$\mathbb{E}[K_t|\mathcal{F}_s] = \mathbb{E}\left[\frac{1}{2}\sum_{i=1}^{n-1}\left(V'_+(x_i) - V'_-(x_i)\right)L_t^{x_i}(X) \middle| \mathcal{F}_s\right] \tag{8}$$
$$= \frac{1}{2}\sum_{i=1}^{n-1}\left(V'_+(x_i) - V'_-(x_i)\right)\mathbb{E}\left[L_t^{x_i}(X)|\mathcal{F}_s\right] \tag{9}$$
$$\leq \frac{1}{2}\sum_{i=1}^{n-1}\left(V'_+(x_i) - V'_-(x_i)\right)L_s^{x_i}(X) = K_s \tag{10}$$

Note that the inequality follows from the fact that $L_t^{x_i}(X)$ is non-decreasing in t, hence $\mathbb{E}\left[L_t^{x_i}(X)|\mathcal{F}_s\right] \geq \mathbb{E}\left[L_s^{x_i}(X)|\mathcal{F}_s\right] = L_s^{x_i}(X)$.

We will consider what happens in different practical cases to these terms, including:

1. A simple one-dimensional SDE with a piecewise linear certificate.
2. A pendulum with a stochastic torque, and X_t measuring the angle of the pendulum.
3. A network b black and w white nodes, with given transition probabilities, and X_t measuring the amount of mass in the white nodes.

to ensure our assumptions are reasonable.

1.1 Proof of the supermartingale property

First, let's note the following:

[1] For example, $\mathbb{E}\left[\int_0^t g(X_s)^2\, ds\right] < \infty$.

Fact

The sum of a martingale and a supermartingale is a supermartingale.

Proof: By definition, given a martingale M_t and a supermartingale S_t, for $r < t$ we have $\mathbb{E}[M_t|\mathcal{F}_r] = M_r$ and $\mathbb{E}[M_t|\mathcal{F}_r] \leq S_r$. Then, for $r < t$:

$$\mathbb{E}[M_t + S_t|\mathcal{F}_r] = \mathbb{E}[M_t|\mathcal{F}_r] + \mathbb{E}[S_t|\mathcal{F}_r] \tag{11}$$
$$= M_r + \mathbb{E}[S_t|\mathcal{F}_r] \tag{12}$$
$$\leq M_r + S_r = (M + S)_r \tag{13}$$

hence $M_t + S_t$ is a supermartingale.

The drift term D_t is a supermartingale by assumption 1. The kink term K_t is a supermartingale by assumption 3. The stochastic integral M_t is a martingale by assumption 2. Therefore, the sum of these three terms is a supermartingale (as a consequence of the previous fact), and hence $V(X_t)$ is a supermartingale.

2 Piecewise $\mathcal{C}^2$ certificates in dimension one

The setup is pretty similiar, with the exception that V is now piecewise $\mathcal{C}^2$ instead of piecewise linear.

Setup

Let:

1. $dX_t = f(X_t)dt + g(X_t)dW_t$ be a one-dimensional SDE.
2. $V(x)$ be a piecewise $\mathcal{C}^2$ function such that $V(x) \in \mathcal{C}^2$, where $x \in (x_i, x_{i+1})$.
3. $V(x)$ is continuous at breakpoints: $V(x_i^+) = V(x_i^-) = V(x_i)$.

We will again consider what conditions on V are sufficient to guarantee that $V(X_t)$ is a supermartingale. Again, the natural hypothesis is:

Hypothesis

Does concavity of $V(x)$ (along with an interior / drift term condition) guarantee that $V(X_t)$ is a supermartingale?

We apply the Itô-Tanaka formula to $V(X_t)$, and we get similiar terms, with the addition of

$$C_t = \frac{1}{2}\int_0^t V''(X_s)g^2(X_s)ds \tag{14}$$

Since the integral in the C_t term is over the same intervals as the D_t term, we can combine them to recover the generator / Lie derivative term:

$$D_t + C_t = \int_0^t \left(V'(X_s)f(X_s) + \frac{1}{2}V''(X_s)g^2(X_s)\right)ds = \int_0^t \mathcal{L}V(X_s)ds \tag{15}$$

The proof that $V(X_t)$ is a supermartingale is similiar to the piecewise linear case, but now we need to impose the condition:

$$\mathcal{L}V(x) = V'(x)f(x) + \frac{1}{2}V''(x)g^2(x) \leq 0 \tag{16}$$

Table 1: Piecewise linear certificates examples.

SDE	$V(x)$	D_t	M_t	K_t
$dX_t = dW_t$	$-\lvert x\rvert$	0	$\int_0^t - \operatorname{sgn}(X_s)\, dW_s$	$-L_t^0(X)$
$dX_t = -X_t\, dt$	$\lvert x\rvert$	$-\int_0^t \lvert X_s\rvert\, ds$	0	0

Table 2: Piecewise $\mathcal{C}^2$ certificates examples.

SDE	$V(x)$	D_t	M_t	C_t	K_t
$dX_t = -\lambda X_t\, dt + \sigma\, dW_t$	x^2	$-2\lambda \int_0^t X_s^2\, ds$	$\int_0^t 2\sigma X_s\, dW_s$	$\sigma^2 t$	0
$dX_t = dt + \sqrt{2}\, dW_t$	$x - \frac{x^2}{2}$	$\int_0^t (1 - X_s)\, ds$	$\int_0^t (1 - X_s)\sqrt{2}\, dW_s$	$-t$	0
$dX_t = dW_t$	$\begin{cases} -\frac{1}{2}x^2, & x < 0, \\ -x, & x \geq 0 \end{cases}$	0	$\int_0^t V'(X_s)\, dW_s$	$-\frac{1}{2}\int_0^t \mathbf{1}_{\{X_s<0\}}\, ds$	$-\frac{1}{2}L_t^0(X)$

3 Studying piecewise certificates on concrete one-dimensional examples

Some useful examples to study are given in Tables 1 and 2.

3.1 Ornstein–Uhlenbeck process in three parameter regimes

Consider the Ornstein–Uhlenbeck process

$$dX_t = -\lambda X_t\, dt + \sigma\, dW_t$$

with the quadratic certificate $V(x) = x^2$. Its generator is

$$\mathcal{L}V(x) = -2\lambda x^2 + \sigma^2 = -2\lambda \left(x^2 - \frac{\sigma^2}{2\lambda} \right).$$

Consequently, if

$$r = \frac{\sigma}{\sqrt{2\lambda}},$$

then the generator is negative for $|x| > r$, zero for $|x| = r$, and positive for $|x| < r$. The decomposition is

$$X_t^2 - X_0^2 = \underbrace{-2\lambda \int_0^t X_s^2\, ds}_{D_t} + \underbrace{2\sigma \int_0^t X_s\, dW_s}_{M_t} + \underbrace{\sigma^2 t}_{C_t},$$

with $K_t = 0$.

Figure 1 fixes $\lambda = 1$ and $X_0 = 1$, and uses $r = 0.5$, $r = 1$, and $r = 1.5$. These choices make the initial generator negative, zero, and positive, respectively. Moreover,

$$\mathbb{E}[X_t^2] = r^2 + (X_0^2 - r^2)e^{-2\lambda t},$$

so the expected generator retains the corresponding sign. In the middle case, X_t^2 is constant in expectation, but it is not a martingale: the pointwise generator vanishes only when $|X_t| = r$.

The stochastic integral M_t need not visibly oscillate on every individual sample path; the martingale statement is instead that its conditional drift is zero. Figure 1 therefore plots Monte Carlo expectations. The mean of M_t remains statistically consistent with zero, while D_t and C_t determine whether $\mathbb{E}[X_t^2 - X_0^2]$ decreases, remains zero, or increases.

3.2 Failure of piecewise linearity and success of curvature

For $f = 1$ and $g = \sqrt{2}$ on $[0, 1]$, consider a case where want to find a certificate V such that $V(0) = 0$ and $V(1) = 1$.

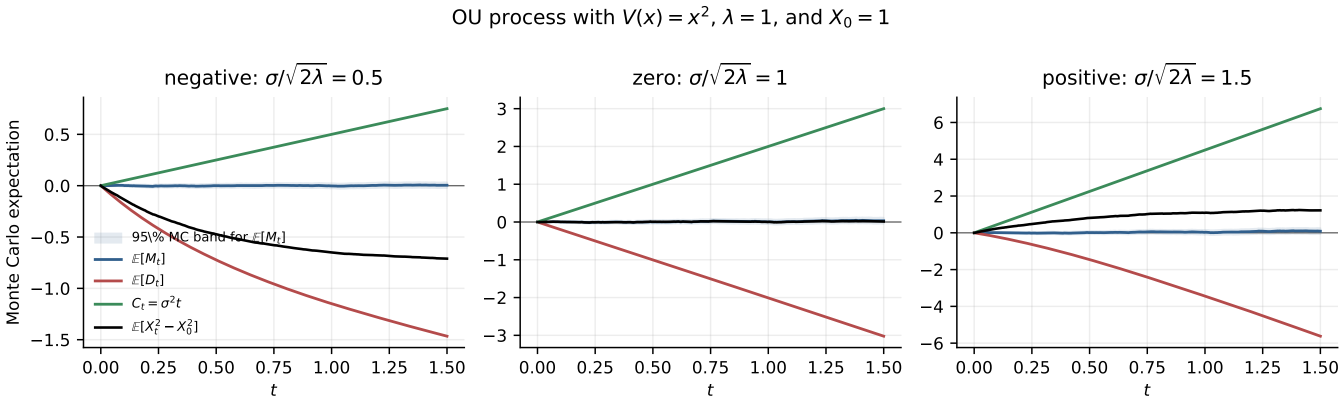

Figure 1: Monte Carlo expectations from 8000 Ornstein–Uhlenbeck paths for $V(x) = x^2$. The three columns use $\sigma/\sqrt{2\lambda} = 0.5, 1, 1.5$. The shaded region is a pointwise approximate 95% Monte Carlo band for $\mathbb{E}[M_t]$.

A smooth candidate could be:

$$V_{\mathcal{C}^2}(x) = x - \frac{x^2}{2},$$

the curvature contribution compensates for the positive drift:

$$\mathcal{L}V_{\mathcal{C}^2}(x) = (1 - x) + \frac{1}{2}(-1)(2) = -x \leq 0.$$

and indeed $V_{\mathcal{C}^2}$ is a valid supermartingale certificate.

On the other hand, let's assume V is piecewise linear. Then there exists an interval $(x_i, x_{i+1}) \subset [0, 1]$ where $V'(x) = a_i > 0$. On that interval, given an $x \in (x_i, x_{i+1})$, the curvature term vanishes, and the generator is

$$\mathcal{L}V(x) = V'(x)f(x) + \frac{1}{2}V''(x)g^2(x) = V'(x)f(x) = V'(x) = a_i > 0.$$

This example motivates the following hypothesis, which we will study in more detail in Section 4.

Hypothesis

While piecewise linear certificates V_{PL} are not sufficient to guarantee that V_{PL} is a supermartingale, **piecewise quadratic certificates** are sufficient, as the curvature term can compensate for positive drift.

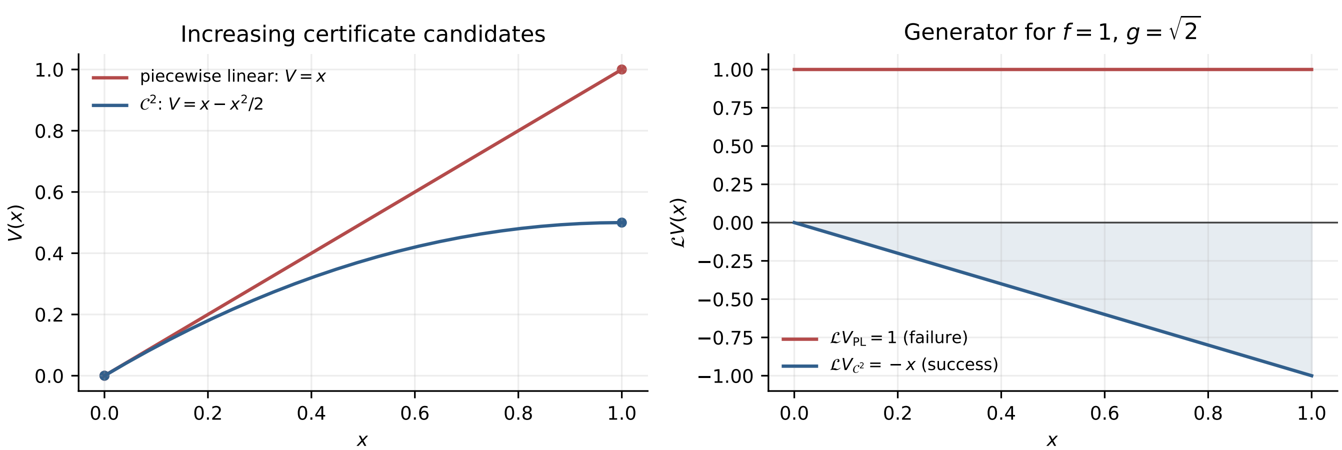

Figure 2: An increasing piecewise-linear candidate fails the generator condition for $f = 1$, $g = \sqrt{2}$, whereas negative curvature makes the smooth candidate satisfy it throughout $[0, 1]$.

3.3 A piecewise $\mathcal{C}^2$ certificate with curvature and a kink

Finally, let $dX_t = dW_t$ and

$$V(x) = \begin{cases} -\frac{1}{2}x^2, & x < 0, \\ -x, & x \geq 0. \end{cases}$$

Here $D_t = 0$, while

$$M_t = \int_0^t V'(X_s)\, dW_s, \qquad C_t = -\frac{1}{2}\int_0^t \mathbf{1}_{\{X_s<0\}}\, ds, \qquad K_t = -\frac{1}{2}L_t^0(X).$$

Thus C_t decreases while the path lies below zero, and K_t decreases only when local time accumulates at the kink. Figure 3 shows these different mechanisms on the same Brownian realization. The plotted local time uses an occupation-density approximation with bandwidth 0.06.

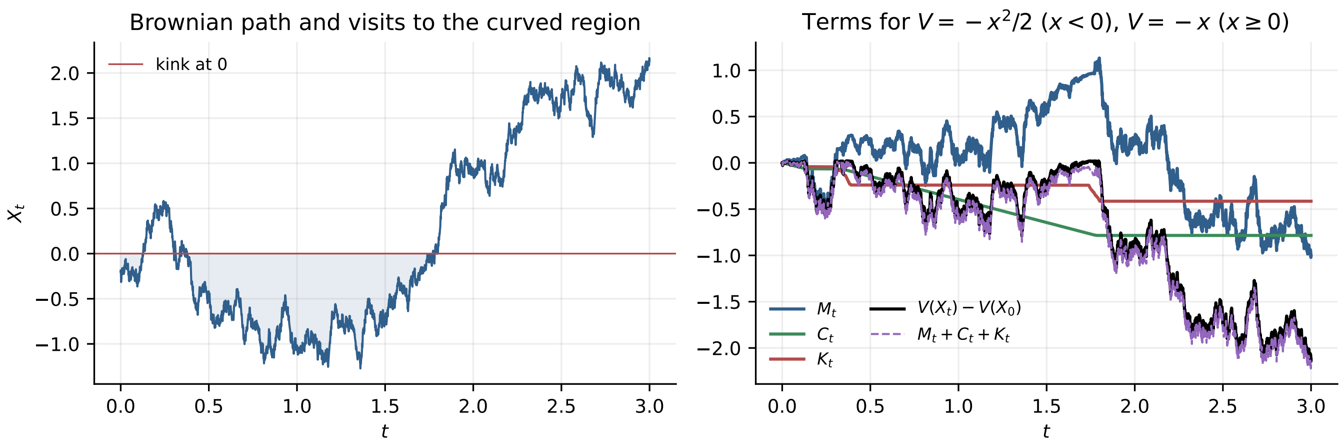

Figure 3: A Brownian realization and the terms for the hybrid piecewise $\mathcal{C}^2$ certificate. The curvature and kink terms are both nonincreasing, while the martingale term fluctuates.

4 Piecewise quadratic certificates in dimension one

In this section, we will study a failure regime of the piecewise linear certificates in dimension one and see how it leads to sufficiency of piecewise quadratic certificates.

Additionally, we observe an interesting connection to the *Taylor expansion* of the best possible certificate, namely a martingale certificate for which the generator is zero.

4.1 Failure of piecewise linear certificates in dimension one

We will consider the setup from Section 1 again, but with additional assumptions on absorption at the boundary, an increase condition on the piecewise linear function $V(x)$, and a non-degeneracy condition on the diffusion term $g(x)$.

The key point is that the piecewise linear setup is not flexible enough when the certificate is required to increase in the same direction as the drift. In that case, a piecewise linear function has no curvature term which could compensate for the positive drift contribution.

So essentially, we will show that:

Fact

(*Informally.*) There is a case where no piecewise linear certificate can make $V(X_t)$ a supermartingale.

Although it should be clear from previous sections, for completeness, the setup is as follows:

Setup

Let:

1. $dX_t = f(X_t)dt + g(X_t)dW_t$ be a one-dimensional SDE.
2. $V(x)$ be a piecewise linear function such that $V(x) = a_i x + b_i$, where $x \in (x_i, x_{i+1})$.
3. $V(x)$ is continuous at breakpoints: $V(x_i^+) = V(x_i^-) = V(x_i)$.
4. as a consequence, $V'_-(x_i) = a_{i-1}$, $V'_+(x_i) = a_i$
5. The X_t process is **absorbed** at the boundary of $I = [a, b]$.
6. $V(x)$ has the **increase condition** at the boundary:

$$V(a) < V(b). \tag{17}$$

7. $g(x)^2 > 0$ for all $x \in (a, b)$, i.e. the **diffusion term is non-degenerate**.
8. M_t is a martingale, i.e. the **martingale condition**.

Motivated by the example given in Section 3.2, we will now show that there is a broad class of SDEs for which increasing piecewise linear certificates cannot be supermartingales.

Fact

Assume that for all $\varepsilon > 0$, there exists $x \in (a, a+\varepsilon)$ such that

$$f(x) > 0. \tag{18}$$

Then there is no continuous piecewise linear V satisfying the increase condition

$$V(a) < V(b) \tag{19}$$

and the concavity condition

$$V'_+(x_i) - V'_-(x_i) \le 0 \tag{20}$$

at every breakpoint, such that $V(X_t)$ is a supermartingale.

Proof. Assume, for contradiction, that such a piecewise linear function V exists and that $V(X_t)$ is a supermartingale.

Let $p(x) = V'(x)$ on each linear region. Since V is piecewise linear, the Itô–Tanaka formula gives

$$V(X_t) = V(X_0) + \int_0^t p(X_s) f(X_s)\, ds \tag{21}$$

$$+ \int_0^t p(X_s) g(X_s)\, dW_s \tag{22}$$

$$+ \frac{1}{2} \sum_i \left(V'_+(x_i) - V'_-(x_i)\right) L_t^{x_i}(X). \tag{23}$$

The stochastic integral is a martingale by assumption. Therefore, if $V(X_t)$ is a supermartingale, the finite-variation part cannot create positive drift on any linear region. Hence the necessary interior condition is

$$p(x) f(x) \le 0 \qquad \text{on every linear region.} \tag{24}$$

This also follows from Fact 1 in Section 1.

We now use the concavity condition. Since

$$V'_+(x_i) - V'_-(x_i) \le 0 \tag{25}$$

at every breakpoint, the slopes of V are nonincreasing from left to right. Therefore V is concave.

Let (a, x_1) be the first linear region. We claim that the slope of V on this first region must be positive. Indeed, suppose instead that the first slope were nonpositive. Since the slopes are nonincreasing, all later slopes would also be nonpositive. Thus,

$$V(b) - V(a) = \int_a^b V'(x)\, dx \tag{26}$$

$$\le 0. \tag{27}$$

This contradicts the increase condition $V(a) < V(b)$. Hence the first slope must be positive, so

$$p(x) > 0 \qquad \text{for all } x \in (a, x_1). \tag{28}$$

Now use the assumption on the drift. Since for every $\varepsilon > 0$ there exists $x \in (a, a+\varepsilon)$ such that $f(x) > 0$, we may choose $\varepsilon = x_1 - a$. Then there exists $x^* \in (a, x_1)$ such that

$$f(x^*) > 0. \tag{29}$$

Combining this with (28), we get

$$p(x^*) f(x^*) > 0. \tag{30}$$

This contradicts the necessary interior condition (24). Therefore no such piecewise linear V can make $V(X_t)$ a supermartingale.

The conclusion is that piecewise linear certificates fail when the certificate is forced to increase in the same direction as the drift. The obstruction comes from the fact that piecewise linear functions have no interior curvature term. Indeed, away from the kinks,

$$\mathcal{L}V(x) = V'(x)f(x), \tag{31}$$

so a positive drift contribution cannot be compensated. This motivates the piecewise quadratic case, where the additional curvature term

$$\frac{1}{2}V''(x)g(x)^2 \tag{32}$$

can offset the positive drift contribution.

4.2 Sufficiency of piecewise quadratic certificates in dimension one

4.3 Connection to Taylor expansion of the best possible certificate